



01 Simplified Role Breakdown (What Each Member Builds)

ROHITH KUMAR — Subsystem A (Flight Control & Pilot)

FLIGHT PILOT

Analogy: Rohith is the "Steering Wheel & Brakes". He makes sure drones take off safely, follow 3D waypoints, and never crash into buildings.

- What to build: C++ node `px4_offboard_node.cpp`.
- Day 1 Goal: Install PX4 SITL & spawn drone.
- How it works: Sends 3D velocity vectors to PX4.
- Success Check: Flies square circuit without crashing.

NIKHIL — Subsystem B (Swarm Network & Tech Lead)

NETWORK ARCHITECT

Analogy: Nikhil is the "Walkie-Talkie & Brain". He connects all drones via Wi-Fi mesh so if 1 drone dies, another automatically takes over leadership.

- What to build: Python `mesh_socket.py` & Gazebo Sim ops.
- Day 1 Goal: Setup Git repo, branches, and ROS 2 bridge.
- How it works: Wi-Fi mesh + Deep JSCC image compression.
- Success Check: Zero signal drop under 60% loss.

VEDANTH SAI RAM — Subsystem C (AI Vision & Target Detector)

AI VISION DETECTOR

Analogy: Vedanth is the "Drone's Eyes". He feeds camera images to YOLOv8 AI to spot disaster victims and pin their exact GPS coordinates.

- What to build: Python `yolo_tensorrt_node.py`.
- Day 1 Goal: Set up PyTorch, CUDA & TensorRT model.
- How it works: Detects targets at 30+ FPS and outputs GPS pins.
- Success Check: Target detection accuracy > 85%.

02 Simplified Role Breakdown (Siva Kesava & Harika)



Subsystem D (Siva Kesava): 3D Mapbox UI Live WebGL Telemetry



Subsystem E (Harika): Audit & Verification Gate G1-G6 Audits

SIVA KESAVA — Subsystem D (3D GIS Dashboard & UI)

UI / UX ARCHITECT

Analogy: Siva is the "Mission Control Screen". He builds the visual 3D satellite map on laptop screens so operators can watch drones in real-time.

- What to build: React component Mapbox3D . jsx.

Day 1 Goal: Create React app & Mapbox token.
- How it works: Receives ROS 2 WebSockets & draws 3D markers.

Success Check: Smooth 3D drone tracking on browser.

HARIKA — Subsystem E (Research, Documentation & Audit Lead)

DOCUMENTATION & AUDIT LEAD

Analogy: Harika is the "Project Manager & Quality Auditor". She handles documentation, logs flight test scores, and creates pitch decks for judges.

- What to build: SUTRA Encyclopedia & Gate Logs.

Day 1 Goal: Init docs/SUTRA_Encyclopedia.md.
- How it works: Tracks test errors and builds presentation slides.

Success Check: 100% complete docs & pitch materials.

03 7-Day Simple Sprint Timeline (Starting Tomorrow, July 28)

Day	Primary Goal	What Each Member Does	Milestone Check
Day 1 (Tomorrow)	Repo & Toolchain Setup	Nikhil inits repo. Rohith sets up PX4 SITL. Vedanth installs TensorRT. Siva sets up React. Harika inits docs.	Clean Build Baseline
Day 2	AI Node Generation	Rohith builds px4_offboard_node.cpp. Nikhil builds mesh_socket.py. Siva builds Mapbox 3D map.	Unit Tests Pass
Day 3	Digital Twin Sim	Launch real_world_digital_twin_swarm.sdf in Gazebo Sim 8. Merge first PR into dev (v0.1-alpha).	Gate G1 Passed ☒
Day 4	Pairwise Integration	Pair 1: Rohith & Nikhil (Flight + Mesh). Pair 2: Vedanth & Siva (Vision + 3D GCS).	Gate G2 Passed ☒
Day 5	Full 5-Subsystem Loop	Run SUTRA_48Hr_Hackathon_Master_Suite.py. Full end-to-end takeoff to victim detection mission flight.	Gates G3-G4 Passed ☒
Day 6	Stress & Failover Test	Inject 60% RF loss, 3.5m/s wind shear, and 30s GPS blackout. Confirm VIO position hold works.	Gates G5-G6 Passed ☒
Day 7	Baseline Freeze & Lock	Merge dev -> main (v1.0-release). Record 2-minute demo video (Sutra_Rescue_Mission.mp4).	100% Locked ☒